Week 3 – Individual – Data Profiling

Simran Amesar - 100007050

1. Dataset Overview & Import

The dataset was imported into Tableau Prep Builder via File connection from **data/week3/tableau_prep_cleaning_exercise.csv.** It contains 100 rows and 18 columns covering order transactions with customer, product, and operational fields.

2. Column inventory

- Identifiers: Order_ID, Customer_ID
- Customer info: Customer_Name, Email, Country, City, Age, Customer_Segment
- Order info: Order_Date, Ship_Date, Product_Category, Quantity
- Financials: Unit_Price, Discount, Sales
- Operational: Payment_Method, Satisfaction_Score, Notes

2. Profiling Results

A Profile step was added in Tableau Prep immediately after the data input to inspect distributions, null counts, data types, and cardinality before any transformation.

2.1 Data quality issues identified

Issue	Description	Severity
Duplicates	5 fully duplicate rows	Critical
Missing values	76 total across 8 columns (max: Satisfaction_Score 15, Payment_Method 14)	High
Date formats	Order_Date has 5+ formats: YYYY/MM/DD, DD/MM/YYYY, MM-DD-YYYY, DD.MM.YYYY, YYYY-MM-DD	High
Case inconsistencies	Customer_Segment: 'CORPORATE', 'Corporate', 'corporate' all present	Medium
Country naming	'Germany', 'germany', 'GERMANY', 'Deutschland', 'Espana' — same country, 5 variants	Medium
Unit_Price formats	Mix of plain numbers, euro-prefix (€215.16), and comma-decimals (9999,99)	Medium
Age outlier	Maximum Age = 130, clearly invalid	Medium
Ship_Date text	'not available' string used instead of null in 2 rows	Low

Product_Category	10 unique values due to case variants (electronics / Electronics / ELECTRONICS)	Low
Sales field	Sales = Quantity x Unit_Price x (1 - Discount) — consistent, no corruption	None

2.2 Missing value detail

Column	Missing	% of Rows	Recommended Action
Satisfaction_Score	15	15%	Exclude from KPIs; leave null
Payment_Method	14	14%	Label as 'Not Specified'
Customer_Segment	12	12%	Impute with mode or label 'Unknown'
Email	10	10%	Leave null (optional contact field)
Notes	10	10%	Leave null (free-text, non-critical)
Age	7	7%	Impute with median (48) or bin as 'Unknown'
Order_Date	5	5%	Exclude rows from time-series analysis
City	3	3%	Leave null or label 'Unknown'

3. Column & Structure Analysis

3.1 Cardinality & identifier analysis

- Order_ID: 95 unique values — cannot serve as a primary key due to 5 duplicate rows. Becomes usable after deduplication.
- Customer_ID: 46 unique values — suitable as a customer-level dimension key.
- Email: only 15 unique values across 90 non-null rows — reused emails, not usable as identifiers.
- Unit_Price: 93 unique values — high cardinality numeric field.
- Low-cardinality fields: Discount (6), Satisfaction_Score (5), Notes (5) — ideal for grouping and filtering.

3.2 Structural integrity checks

- No Product_ID column exists — Product_Category is the sole product dimension.
- Sales = Quantity x Unit_Price x (1 - Discount) verified across all rows — no corruption in the calculated field.
- Ship_Date contains the string 'not available' in 2 rows — must be replaced with null before date parsing.
- After removing 5 duplicates, Order_ID can serve as a reliable record identifier.

4. Cleaning Rules & Transformation Steps

The following cleaning sequence was applied in Tableau Prep. Steps are ordered to avoid dependency conflicts (e.g., Unit_Price must be cleaned before Sales recalculation checks).

#	Cleaning Action	Tableau Prep Method	Expected Outcome
1	Remove duplicates	Remove Duplicates step	100 → 95 rows
2	Replace 'not available'	Replace Values in Ship_Date	2 values → null
3	Fix Unit_Price	Calculated field: strip €, replace comma	All numeric
4	Standardize text case	Group & Replace / Clean step	3+ variants → 1
5	Parse dates	Change Type → Date	String → Date type
6	Fix Age outlier (130)	Calculated field: IF Age > 100 THEN NULL	1 value → null
7	Fill missing Segment	IF ISNULL → 'Unknown'	12 nulls handled
8	Calculated: Days_to_Ship	DATEDIFF('day', Order_Date, Ship_Date)	New column
9	Calculated: Revenue_Band	IF Sales < 100 → Low / Medium / High	New column
10	Validate & export	Output step → .hyper or .csv	Clean dataset

5. Calculated Fields Created

Days_to_Ship

DATEDIFF('day', [Order_Date], [Ship_Date])

Measures operational efficiency. Rows with null dates on either side are excluded from this calculation automatically.

Revenue_Band

IF [Sales] < 100 THEN 'Low' ELSEIF [Sales] < 1000 THEN 'Medium' ELSE 'High' END

Bins the continuous Sales field into three categories for dashboard segmentation.

6. Validation Summary

After running the full cleaning flow, the following checks were performed:

Check	Before Cleaning	After Cleaning

Row count	100	95 (5 duplicates removed)
Duplicate rows	5	0
Order_Date type	String (mixed formats)	Date (YYYY-MM-DD)
Ship_Date 'not available'	2 rows	0 (replaced with null)
Unit_Price type	String (€, comma variants)	Float
Age outlier (>100)	1 row (value: 130)	0 (replaced with null)
Customer_Segment variants	6 (case duplicates)	3 clean + 'Unknown'
Country variants	15 (case/language mix)	Standardized to Title Case
Sales field integrity	Consistent	Consistent (unchanged)

7. ETL vs ELT — Pipeline Context

The final section addresses where profiling fits within a broader data pipeline and which architecture is appropriate for this dataset.

	ETL (this assignment)	ELT
When profiling happens	Before loading (at source)	After loading (inside warehouse)
Tool used	Tableau Prep (this assignment)	SQL, dbt, Snowflake, BigQuery
Data volume	Small to medium files	Large-scale / streaming data
Transformation location	Outside the warehouse	Inside the warehouse
Best for this dataset?	YES — small file, Tableau ecosystem	No — no warehouse needed here

Conclusion

ETL is the appropriate pipeline approach for this dataset. Tableau Prep is an ETL tool: data is profiled and cleaned at the source before being loaded into any analytical system. This assignment followed the correct ETL sequence — profile the raw CSV first, apply transformations in the flow, then export a clean output.

ELT would only be preferable if the dataset were significantly larger (millions of rows), sourced from a live operational database, or destined for a cloud data warehouse where in-warehouse SQL transformations using tools like dbt would offer better scalability and auditability.

8. Screenshots